

Aim:

To implement a variational Autoencoder (VAE) and study its generative ability to reconstruct.

Objective:

1. To understand the concept and working of variational Autoencoder
2. To generate new data points by sampling from a latent distribution
3. To visualize reconstructed and newly generated images.

Pseudocode:

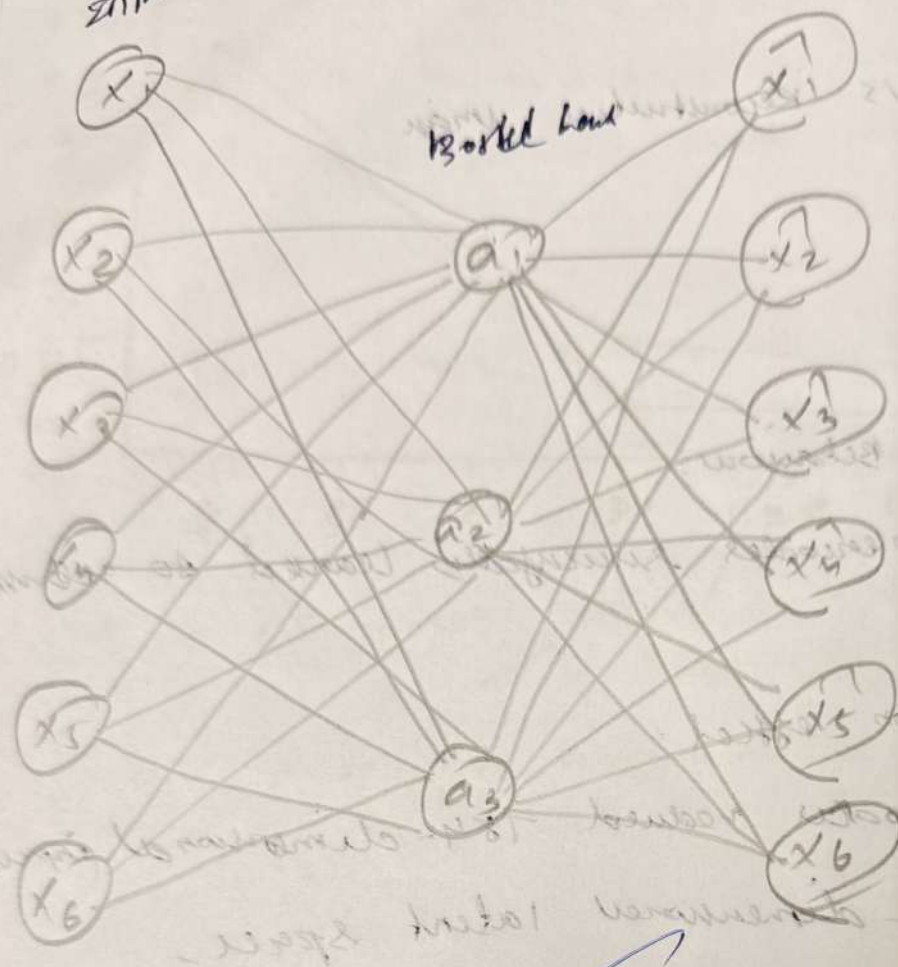
1. Import Libraries (tensorflow, keras, numpy).
2. Load MNIST dataset and normalize image (0-1).
3. Define Encoder:
 - Input Layer (28x28)
 - Dense (256, activation='relu')
4. Sample latent vector z using reparameterization
 $z = \mu + \sigma \cdot \epsilon$, where $\epsilon \sim N(0, 1)$
5. Define Decoder
 - Input: latent vector z
6. Define total loss:

Total loss = Reconstruction loss + KL
7. compile and train model using MNIST dataset
8. visualize:
 - original vs Reconstructed image

Arch
input layer

hidden layer

output layer



3. Visualization
A plot showed strong similarity between original
on reconstructed output
interpretation
The model efficiently performs non-linear dimensionality reduction
in this visualization
Results
The model is highly effective in visualizing the latent space
of the data.

Observations:

1. Training dynamics:

+ The VAE learned to balance between accurate reconstruction and maintaining a continuous latent space.

2. Latent space Representation:

+ Unlike normal autoencoders, the VAE latent space followed a Gaussian distribution.

3. Reconstruction Quality:

+ Reconstructed images were slightly blurrier than those from a normal autoencoder due to stochasticity.

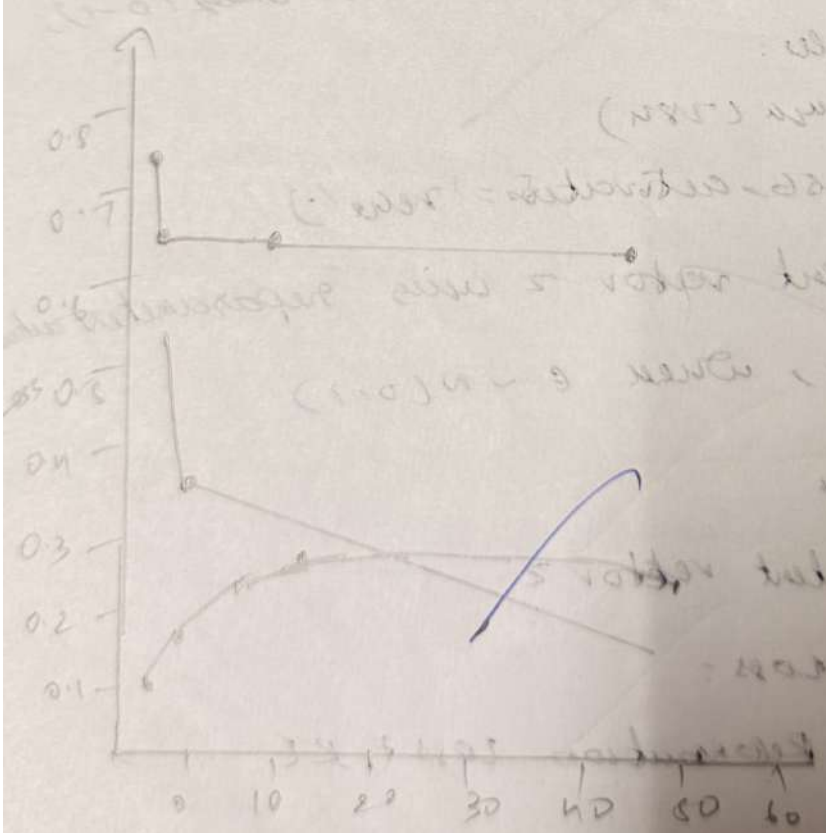
4. Generation Ability:

+ New samples generated by sampling random vectors from latent space.

Result:

Successfully implemented a variational autoencoder (VAE) on the MNIST dataset.

epoch 1 | Recon loss: 148.8329 | total loss: 164.4445
 epoch 2 | Recon loss: 49.2198 | total loss: 121.97
 epoch 3



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